

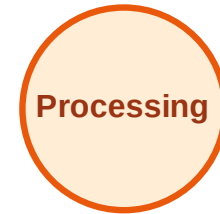
BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Complete



Processing

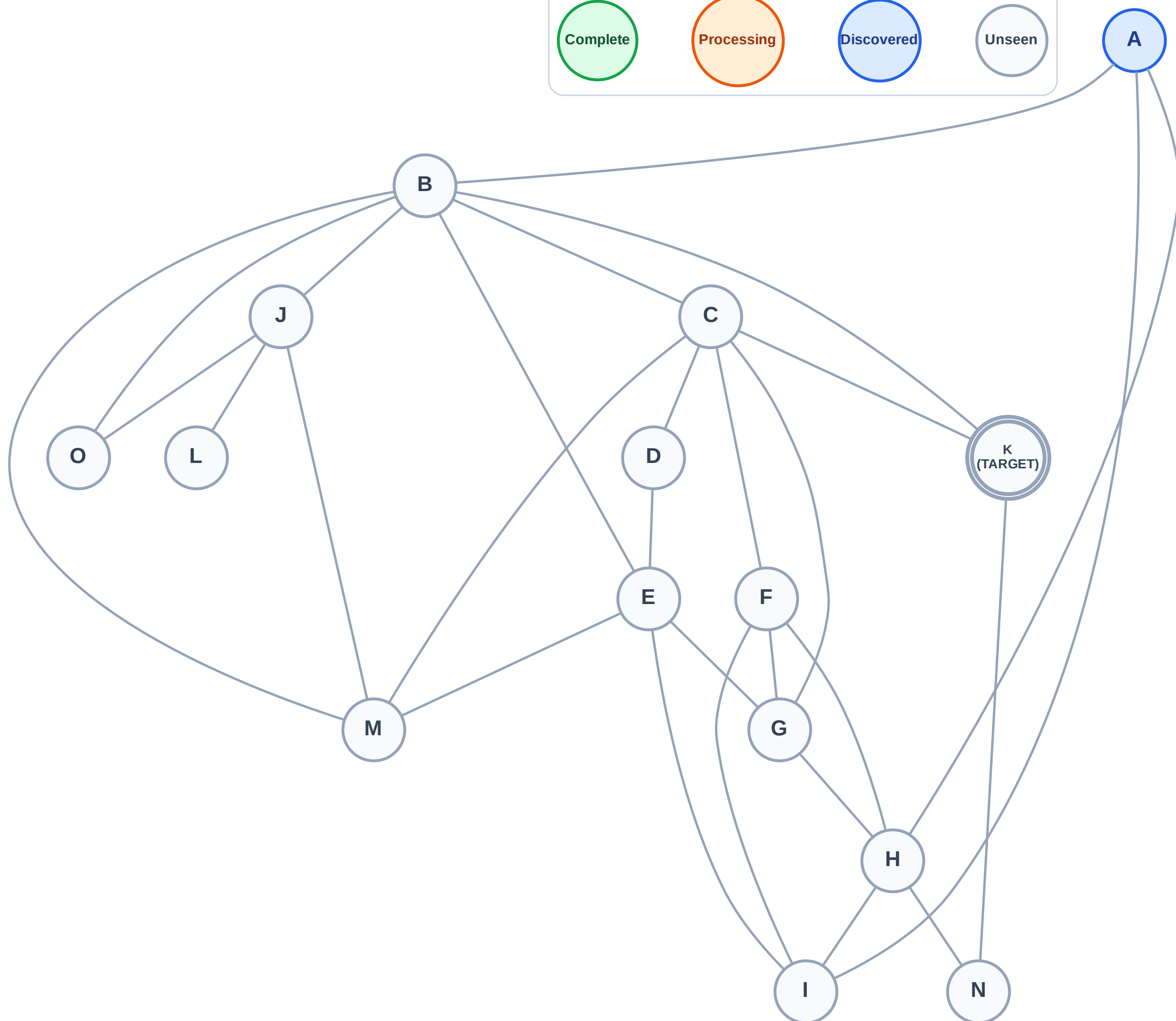


Discovered



Unseen

Queue: A
Visited: (none)



BFS Traversal

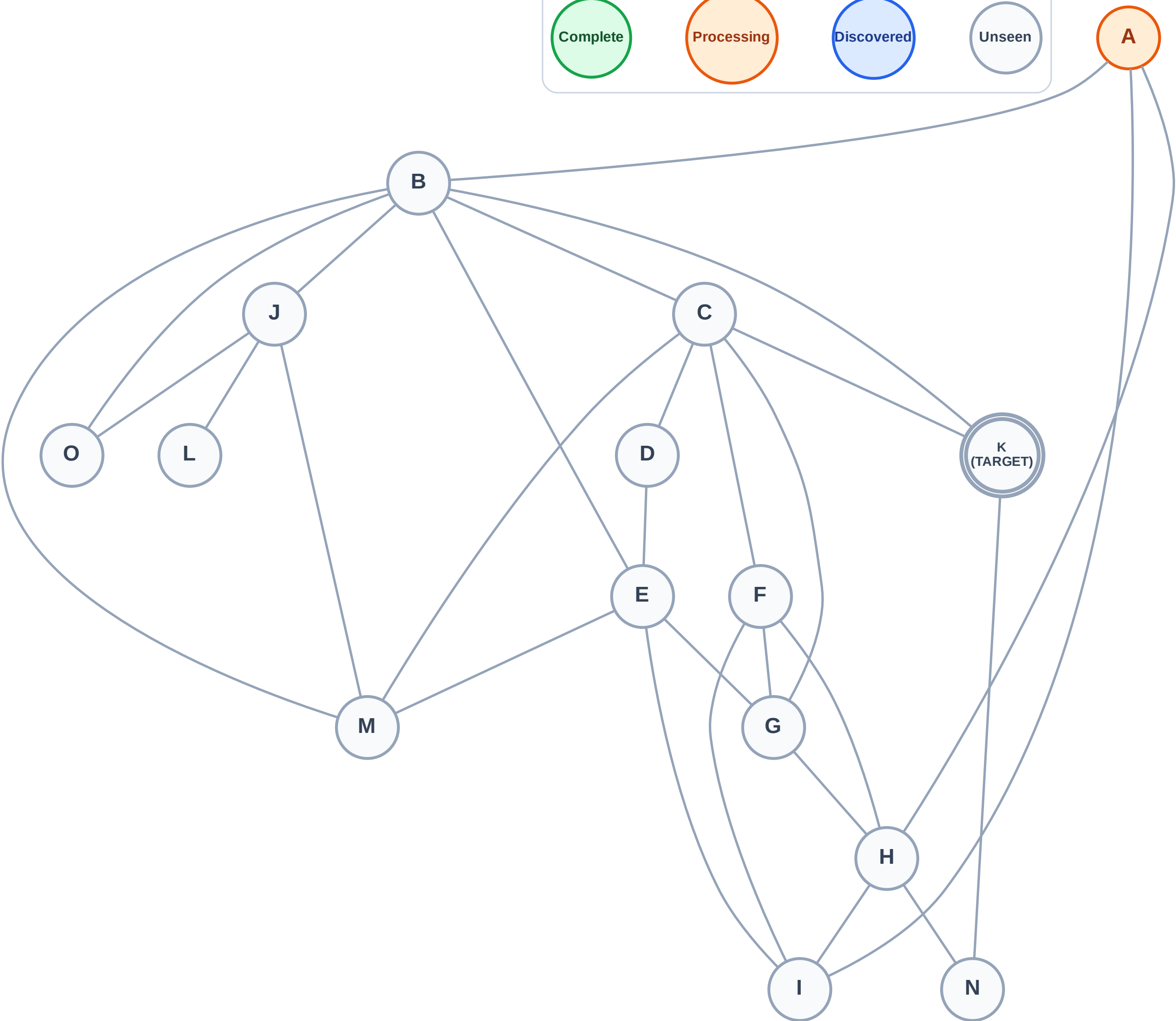
Step 2: Process node A

Legend



Queue: (empty)

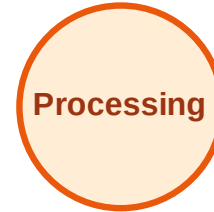
Visited: (none)



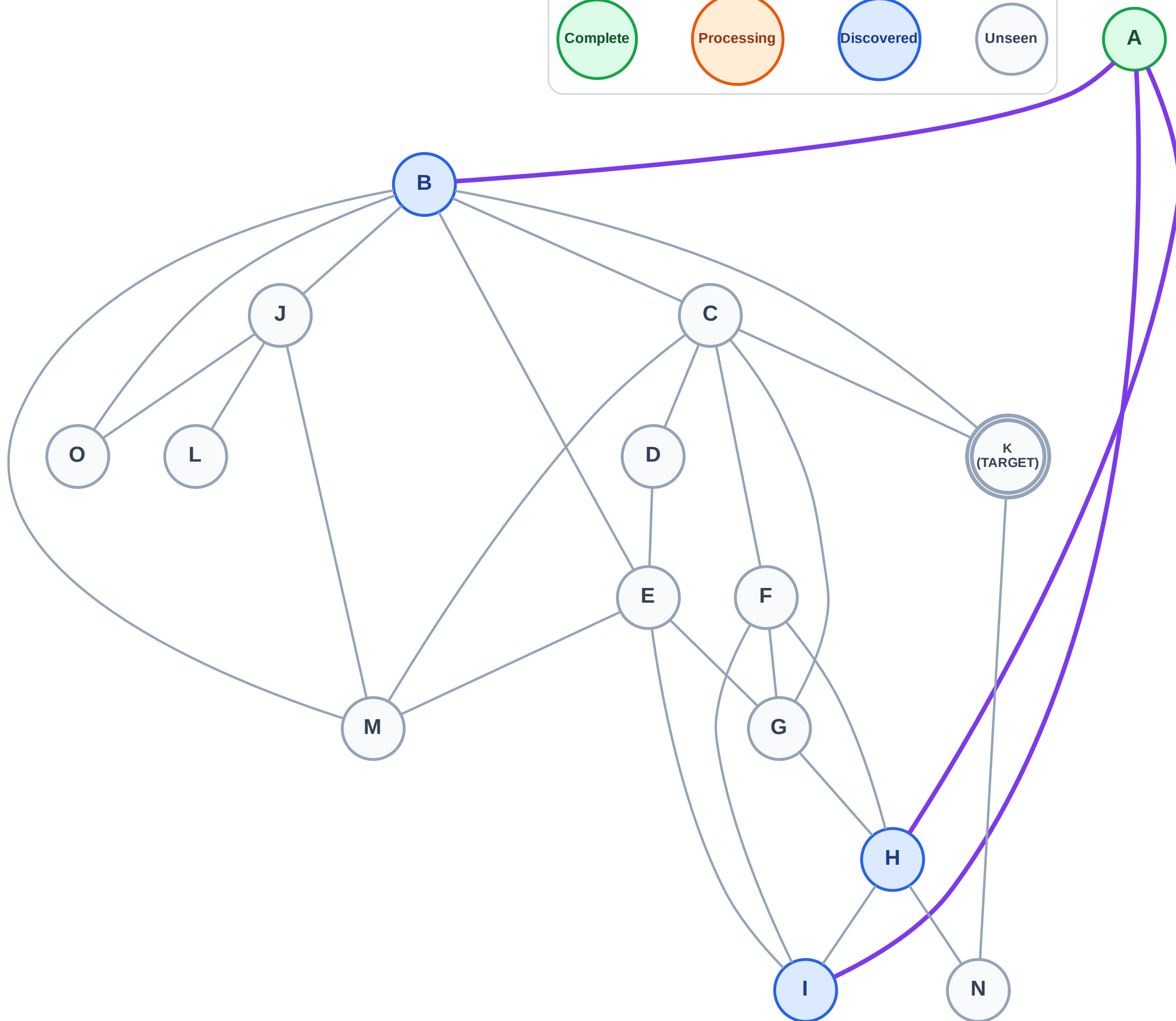
BFS Traversal

Step 3: Discover B, H, I from A

Legend



Queue: B → H → I
Visited: A



BFS Traversal

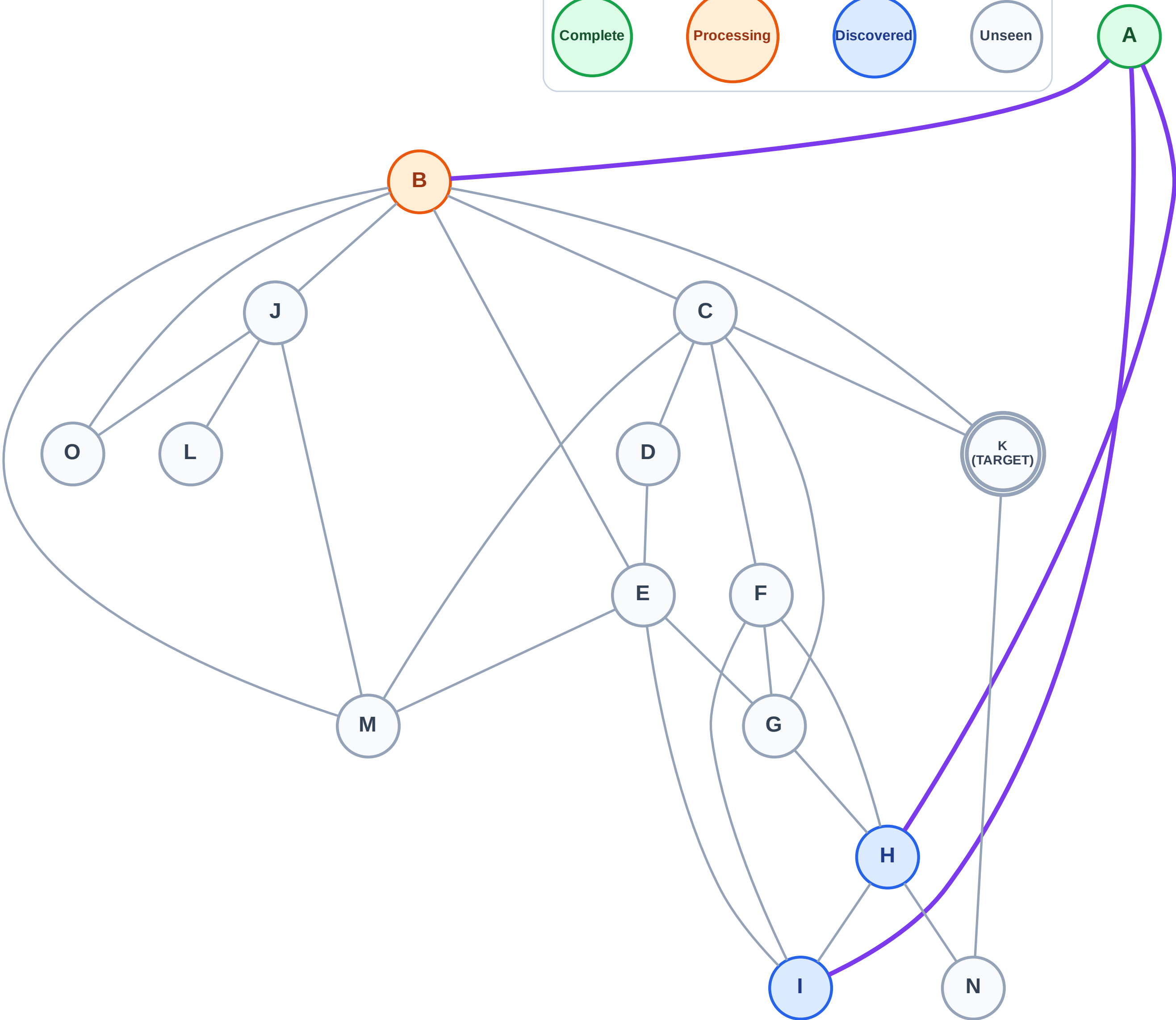
Step 4: Process node B

Legend



Queue: H → I

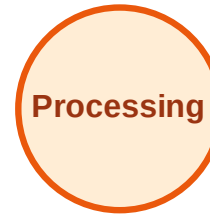
Visited: A



BFS Traversal

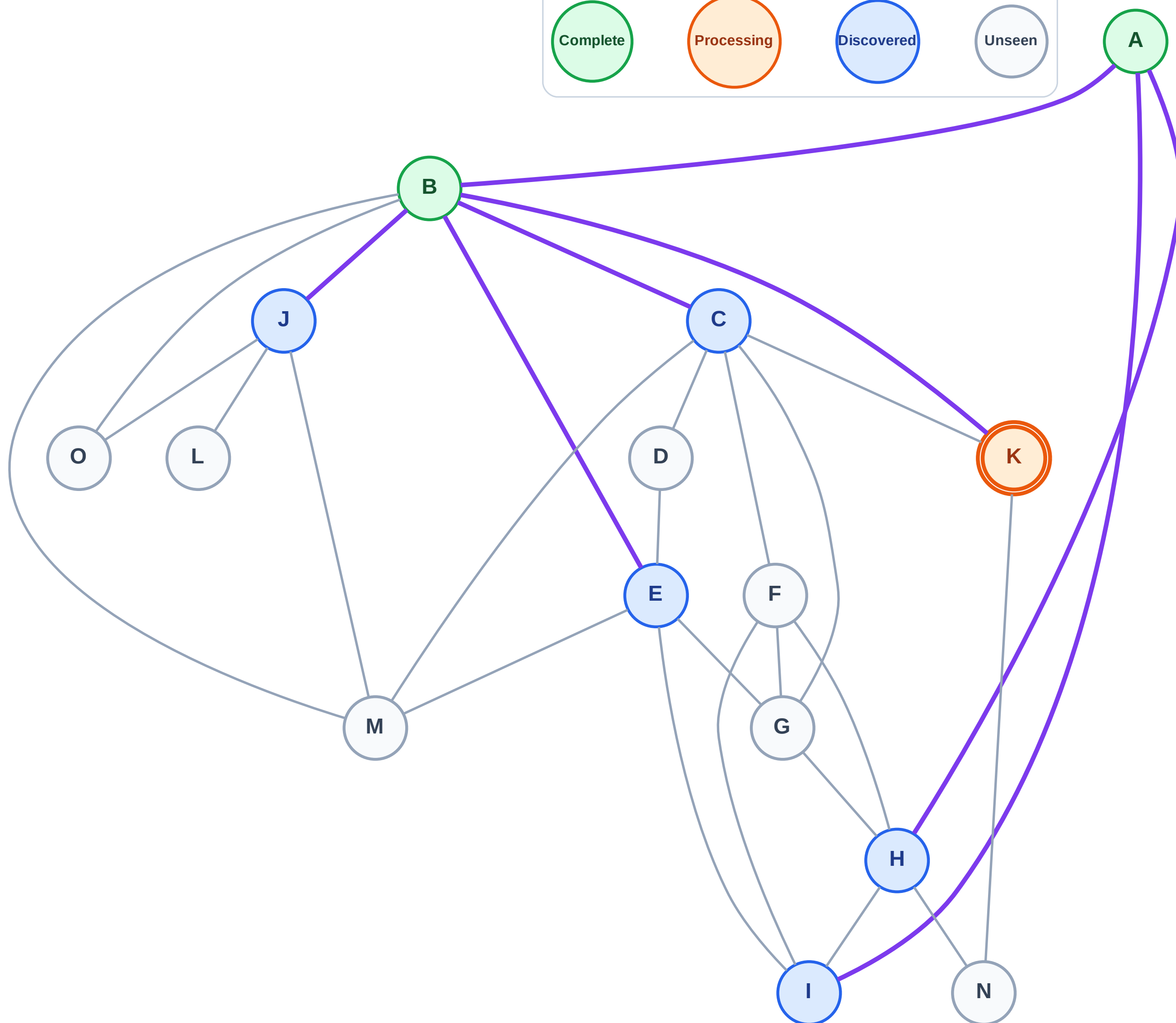
Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → B → K

Legend



Queue: H → I → C → E → J

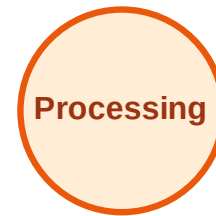
Visited: A, B, K



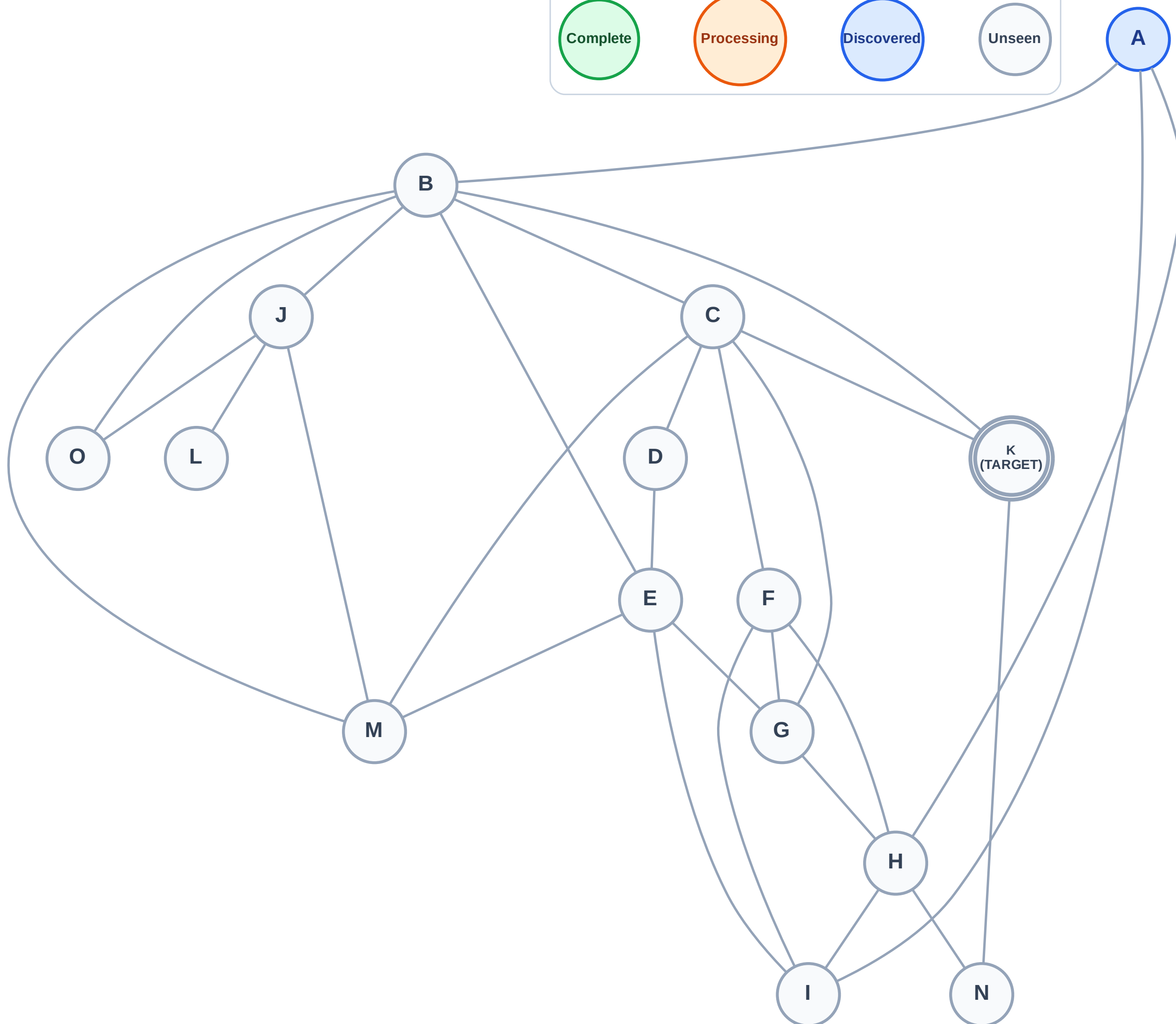
DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

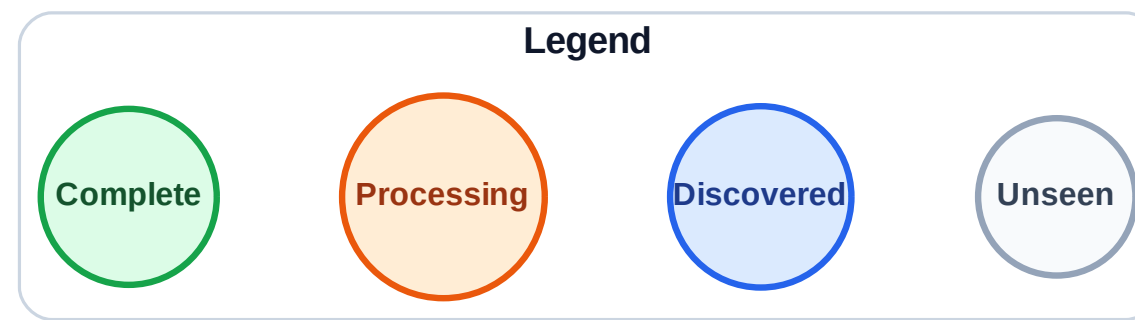


Stack (top at right): A
Visited: (none)



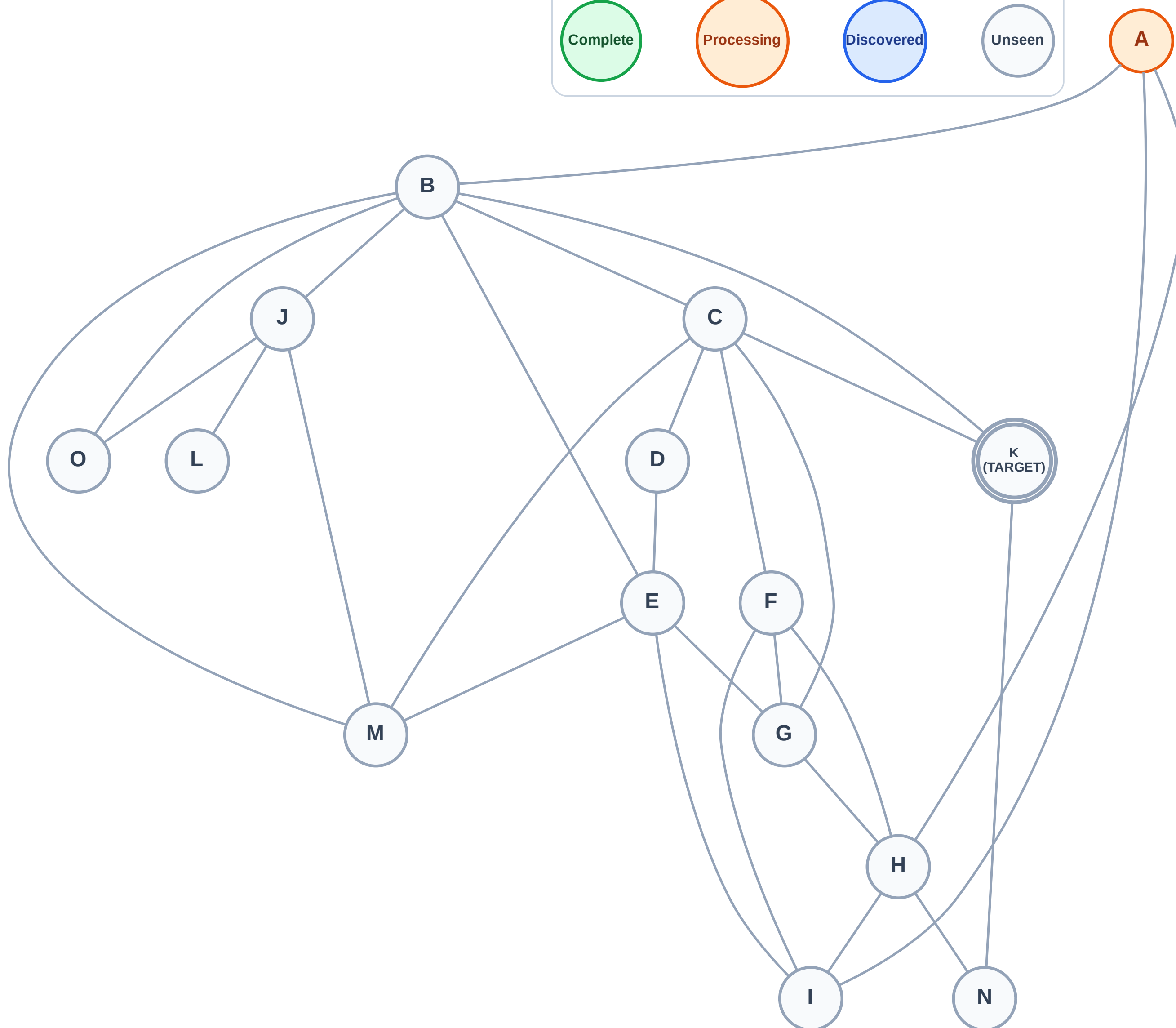
DFS Traversal

Step 2: Process node A



Stack (top at right): (empty)

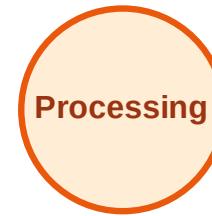
Visited: (none)



DFS Traversal

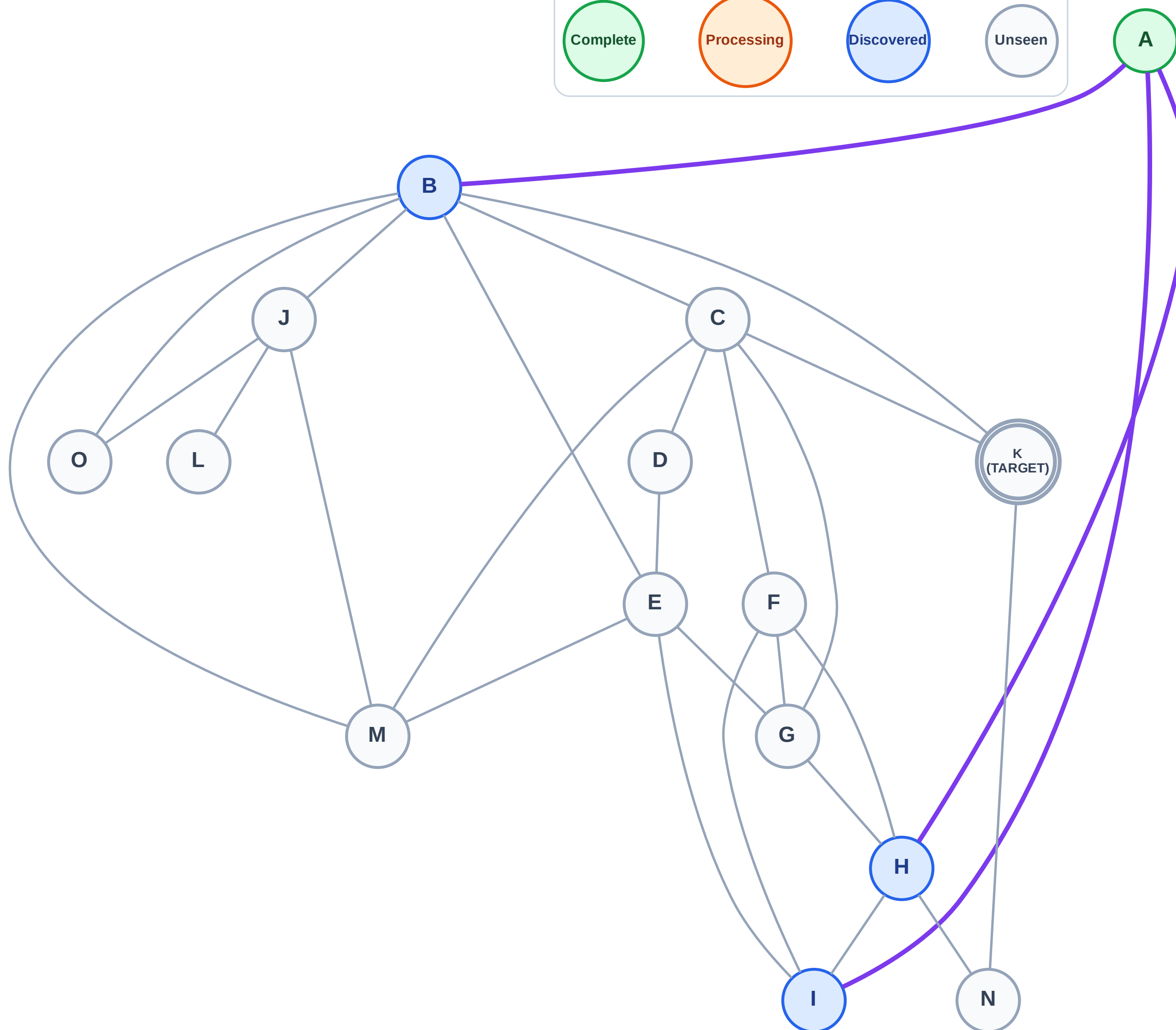
Step 3: Discover I, H, B from A

Legend



Stack (top at right): I | H | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

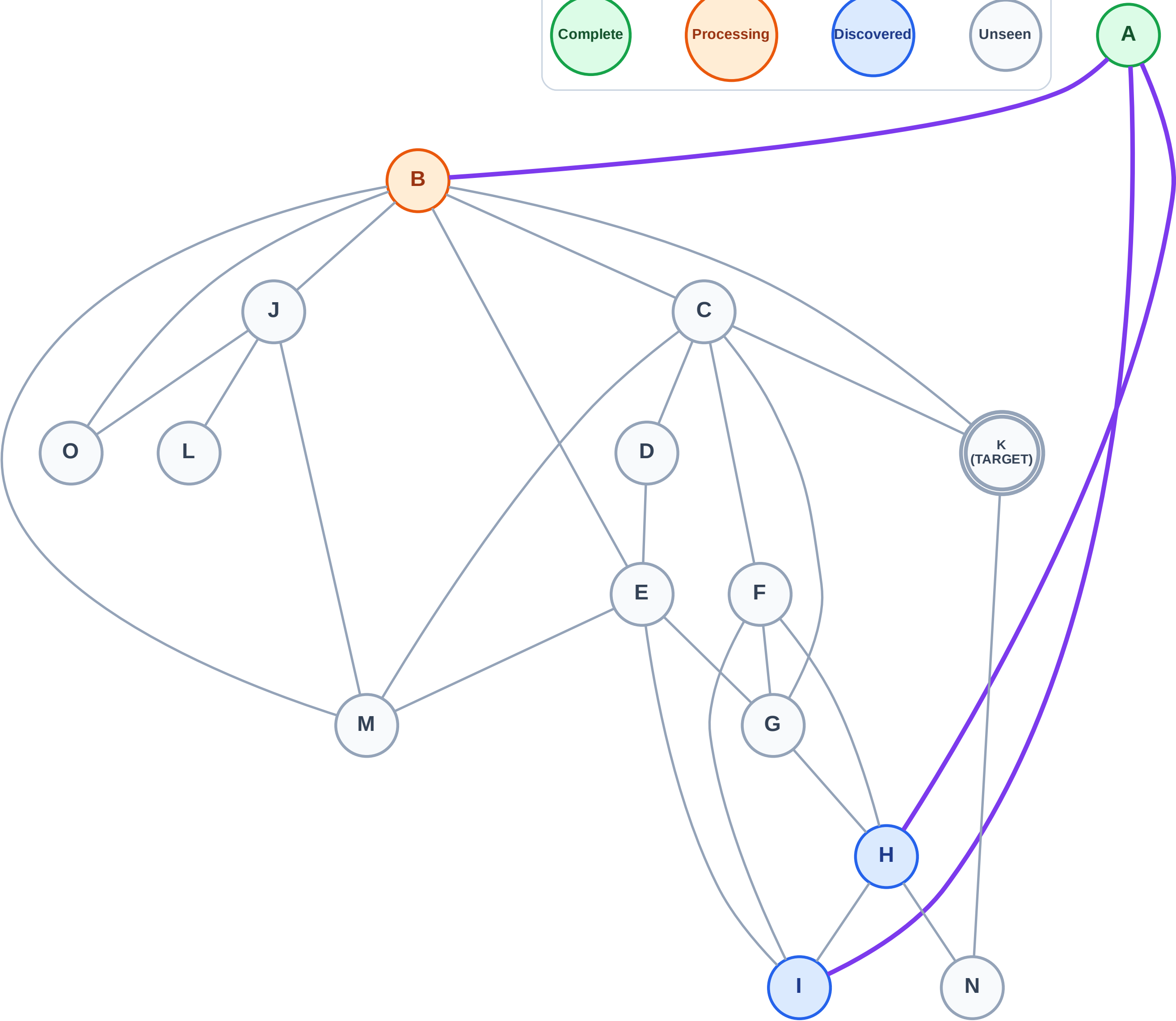
Processing

Discovered

Unseen

Stack (top at right): I | H

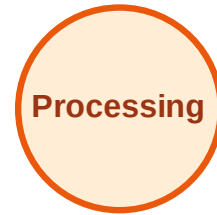
Visited: A



DFS Traversal

Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → B → K

Legend



Stack (top at right): I | H | O | M

Visited: A, B, K

